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Exploring the Foundations of Interplanetary Logistics: Spaceport Infrastructure for the Moon and Mars

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ABSTRACT

*Sustainable human presence on the Moon and Mars requires surface infrastructure capable of supporting landing, refueling, cargo handling, and ISRU. This paper presents a systems architecture for extraterrestrial spaceports, informed by terrestrial logistics analogs (*e.g.*, port operations, ICAO standards) and space mission data (Artemis, Perseverance, VIPER). Key components include regolith-mitigating landing pads, autonomous construction using robotic excavators (*e.g.*, RASSOR), ISRU-to-propulsion chains (*e.g.*, Mars Oxygen In-Situ Resource Utilization Experiment, Sabatier reactors), and modular habitats. The analysis identifies key gaps: limited validation of ISRU-derived materials under lunar/Martian stressors, absence of standardized interface protocols for cross-provider interoperability, and insufficient power solutions for 14-day lunar nights or Martian dust storms. A phased implementation strategy is proposed, from robotic site surveys (2030s) to crew-tended hubs (2040s), prioritizing risk reduction through Earth-based analog testing and international standardization. The framework supports the emergence of an Interplanetary Logistics Grid but does not prescribe governance models. Instead, it offers a technically grounded foundation for infrastructure development aligned with COSPAR planetary protection and sustainability principles.*

Keywords: human space exploration, interplanetary logistics, Mars, moon, spaceport infrastructure, sustainable development

INTRODUCTION

Recent mission planning (*e.g.*, NASA Artemis, ESA Moon Village) and commercial initiatives (*e.g.*, SpaceX Starship) indicate a shift toward sustained off-world operations.[1–3] This transition necessitates the development of extraterrestrial spaceport infrastructure that extends beyond conventional launch and landing functions. Lunar and Martian spaceports are expected to serve as integrated logistical, operational, and industrial nodes, supporting in-situ resource utilization (ISRU), spacecraft servicing, scientific research, habitation, and interplanetary transport.[4] The increasing cadence and complexity of deep-space missions underscore the operational need for robust, scalable infrastructure capable of sustaining regular traffic and enabling permanent human presence.[2] Spaceports will function as vital waypoints in an emerging interplanetary transportation network, facilitating crew rotations, cargo transfer, refueling, and in-space manufacturing. Their implementation is not speculative but a functional prerequisite for long-term mission viability and economic sustainability beyond Low-Earth Orbit (LEO).

Central technical components of this infrastructure incorporate ISRU systems for producing propellant from local resources, for example, water ice on the Moon or atmospheric CO_2 on Mars, thereby reducing reliance on Earth-based resupply[3]; landing and surface infrastructure, including regolith-mitigating pads and thermally stable hangars, designed to withstand extreme environmental conditions; autonomous construction methods, leveraging robotic deployment and additive manufacturing (*e.g.*, 3D printing, microwave sintering) to minimize launch mass and enable on-site adaptability[5]; and communication and navigation networks, supported by cislunar and Mars-orbit relay constellations to manage signal latency and ensure operational continuity.

To coordinate this multinational and multistakeholder effort, this paper proposes the Interplanetary Spaceport Development Authority (ISDA): an international body modeled on the Integrated Network for Commercial Spaceports (INCS) framework.[6] The ISDA would establish technical standards, regulate surface and orbital traffic, oversee resource allocation, and ensure interoperability and equitable access. Its governance model aims to prevent regulatory fragmentation, mitigate jurisdictional conflicts, and institutionalize environmental and safety protocols across celestial bodies. This paper presents a structured approach to the phased development of lunar and Martian spaceports, integrating site selection criteria, technological readiness, and institutional coordination. By anchoring infrastructure planning in empirical data and multilateral cooperation, the proposed framework supports the emergence of a resilient, interoperable interplanetary logistics system.

STRATEGIC SITE SELECTION

The Moon offers a strategically valuable location for the development of interplanetary logistics infrastructure, serving as a potential refueling depot, cargo transfer node, and staging point for deep-space missions. Its proximity to Earth and reduced gravity (approximately 1/6 g) lower the energy requirements for launch and transit, making it a practical testbed for extraterrestrial operations.[7] However, the selection of lunar spaceport sites must be guided by a systematic assessment of interdependent environmental, operational, and engineering constraints.

A primary criterion is the presence of water ice, particularly within permanently shadowed regions at the lunar poles. Spectroscopic and orbital data confirm significant volatile deposits in these areas, which can support life-support functions and enable ISRU for propellant production via electrolysis. The conversion of water into hydrogen and oxygen reduces dependence on Earth-launched consumables and enhances mission sustainability. Consequently, the lunar south pole, specifically the vicinity of Shackleton Crater, emerges as a high-priority candidate due to confirmed ice deposits and favorable terrain.[8]

Complementary to resource availability is access to reliable solar energy. Certain elevated features near the poles, known as “peaks of eternal light,” receive near-continuous illumination owing to the Moon’s low axial tilt ($\sim 1.5°$).[9] This persistent insolation supports uninterrupted power generation for ISRU systems, communications, habitat operations, and autonomous construction. Polar thermal environments are also comparatively stable, avoiding the extreme diurnal temperature swings observed at equatorial latitudes (from −173°C to +127°C),[10] which impose significant thermal stress on materials and electronics.

Topographic and logistical factors further inform site viability. Safe landing and surface mobility require relatively flat, obstacle-free terrain with minimal slope gradients. The Shackleton Crater rim satisfies these criteria and offers scientific value for geological and radiation studies relevant to Mars analog missions. Alternative candidates include Peary Crater at the North Pole and Malapert Mountain, the latter providing continuous line-of-sight communication with Earth and potential for radio astronomy.[11] *Table 1* provides a systematic, multicriteria comparison of the four highest-priority lunar candidate sites, drawing on LRO topographic data,[13] ice-detection results from LCROSS[12] and VIPER,[14] and orbital illumination models. Malapert Mountain emerges as the overall highest-scoring candidate due to its near-continuous solar access and direct Earth line-of-sight, though all four sites present distinct trade-offs between resource proximity and terrain suitability.

On Mars, site selection criteria differ due to the planet’s thin CO_2 atmosphere, greater heliocentric distance, and distinct surface conditions. Solar irradiance is maximized near the equator, where insolation is more consistent despite periodic

Table 1. Candidate Site Comparison of Lunar Spaceport Locations[a]

Criterion	Shackleton Crater Rim	Malapert Mountain	Haworth Crater	Nobile Crater
Solar Illumination (% of lunar day)	~89	~94	~70	~75
Confirmed Water Ice Proximity	High (PSR floor)	Moderate	High	High (LCROSS)[12]
Terrain Slope (°)	<5 (rim)	<3	<7	<4
Earth Visibility (Communications)	Intermittent	Near continuous	Intermittent	Intermittent
Distance to PSR Ice Deposits (km)	<1	3–5	<1	<1
Landing Approach Suitability	Moderate	High	Moderate	Moderate
Seismic/Geological Stability	High	High	Moderate	Moderate
Overall Suitability Score	★★★★☆	★★★★★	★★★☆☆	★★★☆☆

[a]Data compiled from LRO LOLA topography data[13] and Zuber et al.[8]
LOLA, Lunar Orbiter Laser Altimeter; LRO, Lunar Reconnaissance Orbiter; PSR, permanently shadowed region.

dust storms. Equatorial regions also present distinct entry, descent, and landing (EDL) characteristics. Mars' thin atmosphere limits aerodynamic deceleration, increasing reliance on propulsive landing systems for large payloads, though equatorial trajectories benefit from more consistent and predictable atmospheric profiles compared to higher-latitude or polar sites.[15]

Surface accessibility remains important. Viable sites must feature geologically stable, low-relief terrain to support landing, infrastructure deployment, and rover operations. Subsurface water ice, detected via orbital radar and surface missions, represents a key resource for life support and methane-oxygen propellant synthesis via the Sabatier reaction.[16] Regions such as Arcadia Planitia exhibit shallow ice deposits beneath smooth regolith, offering high ISRU potential.[17]

Scientific and operational considerations also influence site selection. Proximity to features like Valles Marineris or Hellas Basin enables geological and climatological research but may introduce logistical complexity.[18] Persistent challenges include dust accumulation on solar arrays and mechanical systems, necessitating mitigation strategies such as electrostatic shielding, vibration-based cleaning, and redundant power systems. Radiation exposure, due to Mars' weak magnetosphere and thin atmosphere, requires habitat shielding via regolith overburden or subsurface placement in lava tubes. Continuous Earth communication will rely on orbital relay networks or strategically positioned ground stations, with equatorial sites offering optimal coverage.[19]

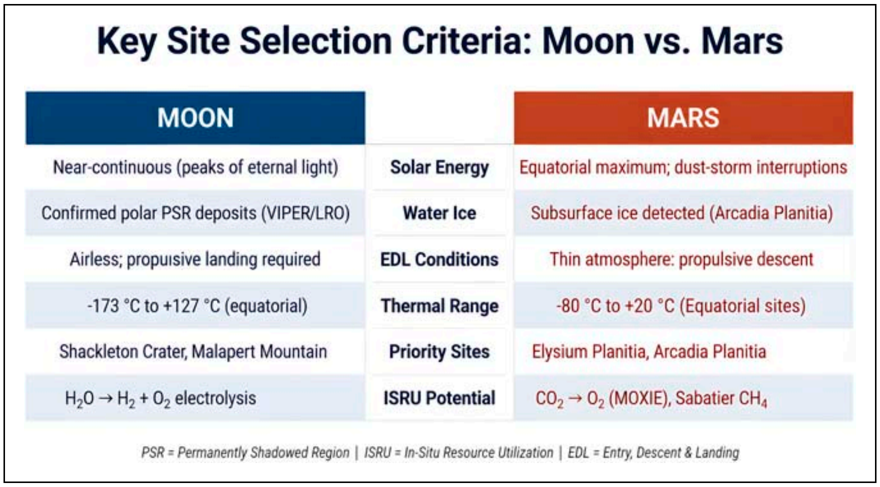
Notable candidates for Martian spaceports include Gale Crater, a well-characterized site explored by NASA's Curiosity rover,[20] which exhibits evidence of past water activity and favorable weather conditions; Elysium Planitia, a broad equatorial plain with flat terrain and excellent solar exposure, already validated by the InSight mission[21]; Arcadia Planitia, which features extensive subsurface ice beneath a smooth surface, making it ideal for ISRU operations; and the flanks of Olympus Mons, which offer elevated terrain and potential access to subsurface resources while serving as a high-altitude staging point for return missions.[17] *Table 2* applies the same comparative framework to four leading Martian candidate sites. Elysium Planitia scores highest overall owing to its equatorial solar irradiance, flat terrain validated by the InSight mission,[21] and moderate ISRU potential, making it the recommended primary anchor site for Phase 2 deployment.

Lunar and Martian spaceports should be developed not as isolated outposts but as integrated nodes within a broader Interplanetary Logistics Grid. Initial deployments will likely prioritize anchor sites at the lunar poles and Martian equator, with phased expansion enabling redundancy and multidirectional traffic. To ensure interoperability, safety, and environmental compliance, the proposed ISDA would establish binding technical standards, coordinate traffic management, and oversee environmental impact assessments. Through this structured, evidence-based approach, spaceport infrastructure can evolve into a resilient and scalable foundation for sustained interplanetary operations. *Figure 1* provides a visual synthesis of the comparative site selection criteria discussed in *Tables 1* and *2*, highlighting the environmental and operational trade-offs between the Moon and Mars across six dimensions: solar energy, water ice availability, EDL conditions, thermal range, priority candidate sites, and ISRU potential.

Table 2. Candidate Site Comparison of Martian Spaceport Locations[a]

Criterion	Elysium Planitia	Arcadia Planitia	Gale Crater	Jezero Crater
Latitude (Equatorial Proximity)	3°N	47°N	5°S	18°N
Solar Irradiance (W/m^2 Avg.)	~180	~130	~185	~175
Subsurface Ice Evidence	Moderate	Strong (SHARAD)[22]	Low	Low
Terrain Flatness	High	High	Moderate	Moderate
Dust Storm Exposure	Moderate	Low-Moderate	Moderate	Moderate
EDL Heritage (Prior Missions)	InSight[21] (2018)	None	Curiosity (2012)[20]	Perseverance (2021)[23]
ISRU Potential (CO_2/H_2O)	Moderate	High	Moderate	Low-Moderate
Overall Suitability Score	★★★★★	★★★★☆	★★★☆☆	★★★☆☆

[a]Data compiled from SHARAD subsurface radar (Bramson et al.),[22] MRO climate observations,[24,25] and NASA Mars Exploration Program site assessments.[26]
EDL, Entry Descent and Landing; ISRU, in-situ resource utilization; MRO, Mars Reconnaissance Orbiter; SHARAD, Mars SHAllow RADar sounder.



Key Site Selection Criteria: Moon vs. Mars

MOON		MARS
Near-continuous (peaks of eternal light)	**Solar Energy**	Equatorial maximum; dust-storm interruptions
Confirmed polar PSR deposits (VIPER/LRO)	**Water Ice**	Subsurface ice detected (Arcadia Planitia)
Airless; propulsive landing required	**EDL Conditions**	Thin atmosphere: propulsive descent
-173 °C to +127 °C (equatorial)	**Thermal Range**	-80 °C to +20 °C (Equatorial sites)
Shackleton Crater, Malapert Mountain	**Priority Sites**	Elysium Planitia, Arcadia Planitia
$H_2O \rightarrow H_2 + O_2$ electrolysis	**ISRU Potential**	$CO_2 \rightarrow O_2$ (MOXIE), Sabatier CH_4

PSR = Permanently Shadowed Region | ISRU = In-Situ Resource Utilization | EDL = Entry, Descent & Landing

Fig. 1. Key Site Selection Criteria for Lunar and Martian Spaceport Locations.

SPACEPORT INFRASTRUCTURE FOR THE MOON AND MARS

Infrastructure Requirements

The development of interplanetary spaceports on the Moon and Mars requires an integrated approach that combines engineering, ISRU, autonomous systems, and international coordination. These facilities must function not only as launch and landing sites but also as multifunctional nodes supporting refueling, maintenance, scientific operations, and habitation. Their design must account for extreme environmental conditions while ensuring interoperability with terrestrial infrastructure and emerging interplanetary logistics networks. Such spaceports are a functional prerequisite for sustained human presence and economic activity beyond LEO.

Current State and Identified Gaps

Current space infrastructure remains limited to mission-specific landing zones with no standardized protocols for refueling, traffic coordination, or long-term operations. While robotic missions have validated the presence of water ice on the Moon and characterized Martian atmospheric and surface conditions, key gaps persist: the absence of operational ISRU-based propellant production systems; no standardized landing pad designs resilient to regolith plume effects; fragmented communication and navigation architectures across agencies; a lack of binding international frameworks for spaceport governance, safety, and environmental protection; and limited validation of autonomous construction and maintenance in extraterrestrial environments. *Table 3* quantifies the current state of readiness for each core spaceport technology, mapping present technology readiness level (TRL) against Phase 1 and Phase 3 targets and identifying the primary remaining validation gap. No technology currently exceeds TRL 7, underscoring the necessity of the phased Earth-analog demonstration strategy outlined in the “Phased Development Approach” section. *Figure 2* illustrates the hub-and-spoke architecture of the proposed extraterrestrial spaceport system, with the ISDA Command Hub coordinating six functional modules: landing pads and hangars, ISRU and fuel production, power utilities, modular habitats, communications and navigation, and autonomous robotics. This architecture is applicable to both lunar and Martian deployments, with module-level adaptations for each body’s environmental conditions.

Fuel Production and Storage

A cornerstone of sustainable interplanetary operations is the ability to produce and store propellant on-site through ISRU. On the Moon, water ice deposits located in permanently shadowed craters at the poles can be extracted using robotic drills, thermal lances, or microwave heating techniques. Once retrieved, this water can be electrolyzed into hydrogen and oxygen, the primary components of high-efficiency rocket fuels.[29] On Mars, the atmosphere, composed predominantly of carbon dioxide, offers a different but equally promising pathway. Technologies such as NASA’s Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) have already demonstrated the feasibility of extracting oxygen from atmospheric CO_2.[30] When combined with hydrogen, either transported from Earth or sourced from subsurface ice, this enables the synthesis of methane fuel through the Sabatier process, establishing a closed-loop propellant production system that drastically reduces dependence on Earth resupply.[31]

Table 3. Technology Readiness Level Assessment of Key Spaceport Technologies[a]

Technology	Current TRL	Target TRL (Phase 1)	Target TRL (Phase 3)	Key Validation Gap
Regolith Microwave Sintering (Landing Pads)	4	6	9	In-situ thermal performance under vacuum cycling
Water Ice Extraction (Lunar PSR)	4	6	9	Drill-to-storage chain at operational scale
H_2/O_2 Electrolysis from Lunar Ice	5	7	9	Cryogenic storage under 1/6 g
MOXIE-Based O_2 Production (Mars)	6	8	9	Continuous operation through dust storms
Sabatier CH_4 Synthesis (Mars)	4	6	9	CO_2 feedstock purity and reactor longevity
Autonomous Robotic Construction	4	6	8	Unstructured terrain adaptability
Kilopower Fission Reactor (Surface)	5	7	9	Long-duration thermal management
Inflatable Habitat Pressure Shells	6	8	9	Micrometeoroid puncture resistance
Delay-Tolerant Networking (DTN)	7	8	9	End-to-end Mars relay latency under a 24-min delay
AI Predictive Maintenance Systems	5	7	9	Fault detection in radiation-degraded hardware

[a]TRL Scale: 1 (Basic Principles) to 9 (Flight-Proven System). Current TRL values based on NASA Technology Taxonomy[27] and ESA TRL Guidelines.[28]
DTN, delay-tolerant networking; TRL, technology readiness level.

To ensure the long-term stability of these cryogenic fuels, spaceports must incorporate advanced storage infrastructure. Propellant tanks will be heavily insulated and equipped with active thermal regulation systems to withstand extreme temperature fluctuations, maintaining liquid oxygen and methane in a stable state for extended periods. Modular refueling stations will support a variety of spacecraft, including crewed vehicles like Starship, cargo haulers, and orbital transfer vehicles. These stations will feature quick-connect interfaces and automated fueling systems, minimize human involvement, and enhance operational efficiency. Autonomous robotic arms and fluid transfer lines will enable rapid, flexible fuel loading without requiring traditional docking, reducing complexity and increasing mission resilience.[32]

Landing Pads and Hangars

The design of landing pads and hangars must account for the harsh realities of extraterrestrial environments. Lunar regolith and Martian dust are fine, abrasive, and electrostatically charged, posing significant risks during landing and surface operations. To mitigate regolith displacement and engine plume damage, landing pads will be constructed from compacted or sintered regolith, forming durable, erosion-resistant surfaces. Emerging concepts such as magnetic suppression fields may also be employed to control dust dispersion, protecting nearby infrastructure and sensitive equipment.[33]

Hangars must provide robust protection against solar radiation, cosmic rays, and micrometeorite impacts.[34] These structures can be built using regolith-covered vaults, lava tube shelters, or composite radiation-shielding materials, creating safe environments for spacecraft storage, maintenance, and repair.[35] Given the extreme thermal cycles (ranging from $-173°C$ to $+127°C$ on the Moon[36] and $-80°C$ to $+20°C$ on Mars[37]), active thermal control systems are essential. Multilayer insulation, phase-change materials, and embedded heating and cooling systems will maintain stable internal temperatures. Surface preparation will be carried out by autonomous rovers equipped with tamping, melting, or 3D printing tools, ensuring stable, reusable landing zones capable of supporting repeated missions.[38]

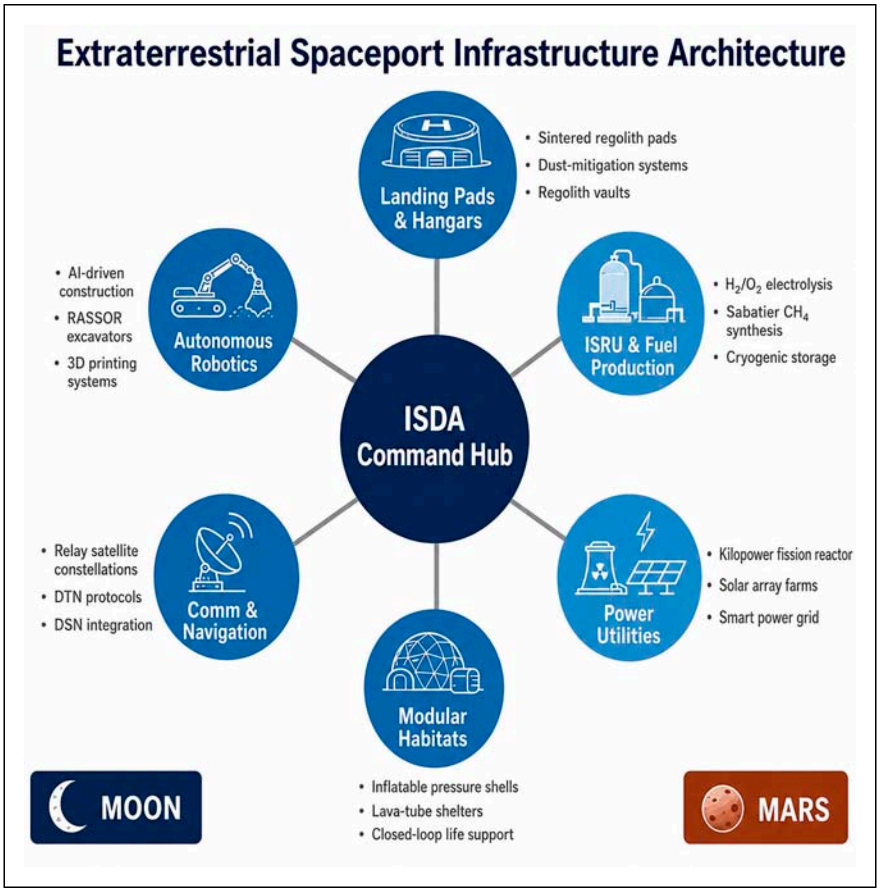


Fig. 2. Extraterrestrial Spaceport Infrastructure Architecture Under the ISDA Command Hub, Applicable to Both Lunar and Martian Deployment.

Communication and Navigation Networks

Reliable communication and precise navigation are vital for coordinating complex interplanetary operations across vast distances. A dedicated constellation of relay satellites orbiting the Moon and Mars will ensure continuous connectivity, eliminate communication blackouts, and enable real-time telemetry and command transmission. These constellations, inspired by missions like the Lunar Communications Relay Demonstration[39] and Mars Reconnaissance Orbiter,[40] will form the backbone of an interplanetary internet.

To manage the inherent delays in deep-space communication (up to 1.3 s between Earth and the Moon[41] and up to 24 min with Mars[42]), Delay-Tolerant Networking (DTN) protocols will be implemented.[43] These systems buffer and forward data packets only when a connection is available, ensuring reliable transmission even during intermittent link conditions. Ground-based navigation beacons and orbiting satellites will provide GPS-like positioning capabilities, using triangulation and time-of-flight calculations to guide landings and surface mobility. This network will be fully integrated with Earth's Deep Space Network (DSN),[44] enabling mission control centers to monitor and direct operations with maximum responsiveness and oversight.

Modular Construction Systems

Rapid, scalable deployment of infrastructure will be enabled through modular construction systems that combine prefabricated components from Earth with in-situ manufacturing. Lightweight, foldable, and inflatable habitats can be launched from Earth and autonomously deployed upon arrival, minimizing launch mass and volume. Autonomous robotic fleets, equipped with manipulators, 3D printers, and construction tools, will handle assembly tasks, reducing the need for human presence in hazardous environments.[45]

The use of local materials is central to sustainability. Lunar regolith can be processed through microwave sintering, molten casting, or binding agents to create durable structural elements such as walls, roads, and blast shields. These additive manufacturing techniques reduce the need for Earth-launched materials and enable on-demand construction.

Habitats and infrastructure will be designed for expandability, allowing new modules to be added or reconfigured based on mission needs, supporting everything from refueling depots and research labs to long-term habitation complexes.[38] The modular construction approach described in this section integrates directly with the six-module spaceport architecture illustrated in *Figure 2*, in which autonomous robotics and ISRU-derived materials underpin all six functional domains.

Interplanetary Spaceport Development Authority

Coordination challenges suggest the potential value of a multilateral forum, conceptually analogous to ICAO, to harmonize technical standards for surface infrastructure.[2,46] This paper proposes the establishment of the ISDA. Modeled after terrestrial frameworks such as the proposed INCS,[6] the ISDA would coordinate the planning, regulation, and operation of lunar and Martian spaceports. Its mission is to enable safe, efficient, and inclusive interplanetary transportation, resource extraction, and human habitation.

The ISDA's vision is to create a sustainable, interconnected interplanetary transportation network, with the Moon and Mars as strategic hubs that facilitate the growth of the space economy and humanity's enduring presence beyond Earth. To achieve this, the ISDA would comprise specialized departments: Infrastructure Development for designing and constructing landing pads, habitats, and processing facilities; Transport and Logistics for managing cargo, fuel, and personnel; Space Traffic Management for scheduling, collision avoidance, and orbital slot allocation; International Relations and Policy for diplomatic engagement and treaty coordination; Resource Management for regulating the use of lunar and Martian materials; and Research and Development (R&D) for advancing technologies in robotics, ISRU, and propulsion.

To govern spaceport activities effectively, the ISDA will establish a comprehensive legal and regulatory framework aligned with international space law. This includes strict adherence to the Outer Space Treaty's principles of peaceful use and nonappropriation,[47] collaboration with the United Nations Office for Outer Space Affairs (UNOOSA)[48] and COPUOS,[49] and alignment with international space law to promote equitable resource use. Environmental protection protocols will safeguard scientifically valuable sites, minimize contamination, and manage waste. Safety and operational standards will standardize procedures for landings, refueling, and crew operations, while space traffic rules will regulate orbital paths and emergency protocols to prevent congestion and collisions.

The ISDA will oversee the design and deployment of key infrastructure, including specialized landing and launch pads, cargo handling and storage facilities, refueling stations, habitat construction zones, and resource processing plants. Each facility will be modular, scalable, and resilient, with built-in redundancy and adaptability for future upgrades. Public–private partnerships (PPPs) will drive investment, while multilateral agreements will define usage rights, responsibilities, and benefit-sharing mechanisms. Funding will be sourced from government grants, private investment, launch fees, and resource extraction royalties.

Risk mitigation and continuous improvement will be central to the ISDA's mandate. Comprehensive risk management strategies will address radiation exposure, micrometeoroid impacts, and Martian dust storms. Systematic assessments of infrastructure integrity and technology performance will be conducted regularly, supported by real-time monitoring tools powered by artificial intelligence (AI) and sensor networks for predictive maintenance and anomaly detection. A centralized spaceport health dashboard will provide member states and partners with live status updates and alerts.

Human capital development will be prioritized through specialized training programs for astronauts, engineers, and technicians in low-gravity construction, life-support systems, and autonomous robotics. STEM outreach and capacity-building initiatives will target emerging spacefaring nations, while cross-disciplinary exchange programs will foster knowledge transfer through internships, joint missions, and simulations. The ISDA will partner with universities to develop curricula and certification programs for off-world engineering and logistics professionals.

The implementation of the ISDA will follow a phased strategy inspired by the proposed INCS model.[6] Phase 1 will focus on consortium building, feasibility studies, and securing funding through PPPs. Phase 2 will establish pilot nodes on the Moon, linked with terrestrial spaceports such as the proposed Omega Spaceport in Kenya,[50] to test interoperability, logistics, and governance. Phase 3 will expand the network and deploy AI-driven scheduling and blockchain-based transaction systems to optimize operations. Phase 4 will finalize the ISDA's governance structure and align its policies with COPUOS to ensure global equity and sustainability.[49] *Table 4* provides order-of-magnitude mass and peak power estimates for each core infrastructure module for both lunar and Martian deployments. The aggregated Phase 1 lunar manifest of approximately 8,550 kg corresponds to six to eight heavy-lift launches, establishing a realistic baseline for procurement and logistics planning.

Table 4. Infrastructure Mass and Power Budget Estimates[a]				
Infrastructure Module	Estimated Mass (kg)—Lunar	Estimated Mass (kg)—Mars	Peak Power Draw (kWe)	Launch Vehicle Manifest
Landing Pad Structure (Per Pad)	2,000	1,500	–	Multiple launches
ISRU Electrolysis Unit	500	450	8–12	Single heavy-lift
Cryogenic Propellant Storage (1 Ton Cap.)	800	750	2–4	Single heavy-lift
Kilopower Fission Reactor (10 kWe)	1,500	1,500	–(generates)	Single heavy-lift
Solar Array (10 kWe Supplemental)	200	350	–(generates)	Rideshare
Inflatable Habitat Module (4-Person)	3,000	2,800	5–8	Single heavy-lift
Communications Relay Node (Surface)	150	150	0.5–1	Rideshare
Autonomous Construction Rover	400	400	1–3	Single medium-lift
Estimated Phase 1 Total (Lunar)	~8,550	–	~17–28	~6–8 heavy-lift launches
Estimated Phase 2 Total (Martian)	–	~7,900	~17–28	~6–8 heavy-lift launches

[a]Mass estimates are order-of-magnitude projections based on analogous NASA[51] and ESA[52] system designs.[35,53] Actual values subject to detailed systems engineering analysis. kWe, kilowatts electric; ISRU, in-situ resource utilization.

Future Directions

The ISDA will evolve to support new spaceports across the Moon and Mars, with future plans for outposts in the asteroid belt and beyond. These hubs will serve as interplanetary trade centers for water, fuel, and rare metals, while also supporting space tourism and early human settlements. The ISDA will coordinate with concepts like the proposed Asteroid Belt Network (ABN) to establish refueling and manufacturing hubs between Mars and Jupiter.[54] Unified command and control systems will provide real-time oversight of spacecraft, resources, and missions, while open-data policies and citizen science initiatives will engage the global public in mapping, analyzing, and designing the future of space. By integrating technical rigor, regulatory coherence, and equitable access into its foundational architecture, the ISDA offers a pragmatic pathway toward a resilient and coordinated interplanetary transportation system.

CHALLENGES AND MITIGATION STRATEGIES

Challenge Context

The establishment and operation of spaceports on the Moon and Mars entail significant environmental, physiological, and engineering challenges that directly affect mission safety, infrastructure integrity, and operational continuity. These extraterrestrial environments are characterized by high radiation flux, reduced gravity, thermal extremes, abrasive regolith, and intermittent energy availability. Addressing these hazards requires a systems-based approach that integrates materials science, autonomous operations, human factors engineering, and robust operational protocols to support sustained human presence and logistical functionality.

Radiation Exposure

One of the key hazards in deep space is exposure to ionizing radiation, including galactic cosmic rays (GCRs), solar particle events (SPEs), and secondary radiation generated when high-energy particles interact with materials.[55] On the Moon, the absence of both a protective atmosphere and a global magnetic field leaves the surface fully exposed to this radiation, making long-duration missions particularly hazardous.[56] Without adequate shielding, astronauts face elevated risks of cancer, central nervous system damage, and degenerative tissue effects, while sensitive electronics are vulnerable to single-event upsets and cumulative degradation.[57]

Mitigation strategies focus on passive and active shielding. One of the most effective approaches is the use of lunar regolith as a natural radiation shield.[58] By piling regolith over surface habitats or using it in construction through sintering or regolith-concrete composites, significant radiation attenuation can be achieved. Alternatively, subsurface habitats (excavated within lava tubes or purpose-built underground chambers) offer exceptional protection from both radiation and micrometeorite impacts, while also providing greater thermal stability.[59] For surface or temporary installations, deployable radiation barriers made from lightweight, inflatable structures lined with polyethylene or hydrogen-rich materials can provide portable shielding. Looking further ahead, experimental concepts such as portable magnetic field augmentation, inspired by Earth's magnetosphere, may one day offer active deflection of charged particles, though such technologies remain in the research phase.[60]

On Mars, the situation is somewhat improved but still severe. The planet's thin atmosphere (about 1% the density of Earth's[61]) provides limited attenuation, particularly against SPEs, but offers little protection from GCRs.[62] Surface radiation levels are lower than on the Moon but remain a long-term threat due to the extended duration of Mars missions. Similar to lunar strategies, subsurface bases built within lava tubes or buried habitats are among the most promising solutions. In addition, water-layered shields, where water is stored in walls or surrounding living quarters, leverage water's high hydrogen content to effectively block radiation. The use of radiation-resistant materials such as high-density polyethylene, borated plastics, and hydrogen-rich composites in habitat construction further enhances shielding capabilities. For future missions, active magnetic field generators using superconducting coils could create localized protective bubbles, offering a dynamic defense against charged particles.[63]

Operational protocols will include continuous radiation monitoring via wearable dosimeters and fixed sensors, with automated alerts triggering shelter-in-place procedures during SPEs. These data will inform habitat design standards and mission duration limits to ensure compliance with permissible exposure thresholds.[64]

Microgravity Effects

Prolonged exposure to reduced gravity environments (approximately one-sixth of Earth's gravity on the Moon[7] and one-third on Mars[24]) poses significant physiological and mechanical challenges. For humans, the lack of gravitational loading leads to rapid muscle atrophy and bone density loss, increasing the risk of fractures and long-term health complications. Cardiovascular deconditioning occurs as blood redistributes toward the upper body, affecting heart function and fluid balance. In addition, vestibular disruption alters spatial orientation and coordination, impairing performance in complex tasks such as piloting, robotic operations, or emergency response.[65]

To counteract these effects, a combination of medical, engineering, and operational countermeasures will be required. Artificial gravity simulators, such as rotating centrifuge modules or full-scale rotating habitats, could provide periodic exposure to Earth-like gravity, helping to maintain musculoskeletal and cardiovascular health. In the near term, structured physical conditioning programs using resistance machines, treadmills with harnesses, and resistance bands will be essential for preserving muscle and bone integrity. These regimens must be integrated into daily routines to ensure compliance and effectiveness.[66,67]

For robotic and mechanical systems, microgravity affects fluid dynamics, lubrication, dust adhesion, and mobility. Adaptive robotics will be necessary, equipped with enhanced traction mechanisms, force feedback control systems, and low-gravity-optimized mobility platforms. Human factors engineering will play a key role in designing user interfaces, tool ergonomics, and movement aids such as handrails, harnesses, and exoskeletons to support safe and efficient operations in these environments.[68]

Energy Sustainability

Energy availability is a cornerstone of spaceport functionality, and both the Moon and Mars present distinct challenges in generation, storage, and distribution. On the Moon, lunar polar regions offer a unique advantage: certain elevated areas receive near-continuous sunlight, enabling near-perpetual solar power harvesting. This makes them ideal locations for powering ISRU facilities, communication arrays, and life-support systems. However, equatorial regions experience a 14-day lunar night, during which temperatures plummet below −173°C,[36] necessitating robust energy storage or alternative power sources.[69]

On Mars, solar irradiance is about 44% of Earth's, and while equatorial zones receive consistent sunlight, dust storms can last for weeks and severely reduce solar panel efficiency.[70] To address this, dust-resistant coatings and self-cleaning surfaces, using vibration, electrostatic discharge, or UV illumination, are being developed to maintain panel performance. Solar tracking systems with deployable arrays can maximize energy capture by following the Sun's path across the sky.[71]

A hybrid power architecture is essential for reliability. This includes the integration of compact nuclear fission reactors, such as NASA's Kilopower project,[72] which can generate up to 10 kilowatts of continuous power, which is sufficient to support multiple habitats and industrial operations. These reactors provide baseload power for life-support systems, fuel production, and scientific instruments, and serve as emergency backup during lunar eclipses or Martian dust storms. Energy storage solutions include advanced battery systems (*e.g.*, lithium-ion or solid-state), molten salt thermal storage for process heat, and hydrogen fuel cells that convert stored hydrogen and oxygen back into electricity and water.[73] An integrated, smart power grid will be required to manage load balancing, prioritize important systems, and enable autonomous diagnostics and maintenance across the spaceport.

Additional Environmental and Operational Challenges

Beyond radiation, gravity, and energy, a range of other environmental and operational challenges must be addressed. Lunar regolith is fine, abrasive, and electrostatically charged, posing risks to seals, joints, and astronaut suits. Martian dust is similarly problematic, adhering to surfaces and degrading solar panels and sensitive electronics. Mitigation strategies include nanocomposite dust-resistant coatings, improved sealing mechanisms, and electrostatic dust suppression systems that neutralize static charge. Techniques such as microwave sintering can stabilize regolith for use in landing pads and roadways, while airlock cleaning stations and vacuum-based systems can remove dust before entry into habitats.[74]

Communication and navigation delays are another major constraint. The one-way signal delay between Earth and the Moon is approximately 1.28 s,[41] while on Mars, it ranges from 4 to 24 min depending on orbital alignment.[42] This makes real-time control from Earth impractical, necessitating autonomous decision-making for navigation, refueling, and emergency response. DTN protocols allow data to be stored and forwarded when connections are available, ensuring reliable communication. Relay satellite constellations around the Moon and Mars will provide continuous coverage, while local AI-based control centers will manage vehicle traffic, system diagnostics, and emergency protocols without Earth intervention.

Both celestial bodies experience extreme thermal cycles. The Moon swings from −173°C at night to +127°C during the day,[36] while Mars averages −80°C[37] but can drop to −125°C at the poles.[75] These fluctuations cause material fatigue, thermal stress, and electronic performance degradation. Mitigation includes multilayer insulation (MLI), phase-change materials, and vacuum-insulated panels, as well as passive radiators, heat pipes, and active climate control systems. Site selection will also play a role, with thermally stable locations, such as permanently lit zones on the Moon or subsurface shelters on Mars, being prioritized.[76]

On Mars, global dust storms can last for weeks, reducing visibility, impairing solar power, and altering atmospheric dynamics during EDL. Storm prediction models using orbital and surface meteorological data will be important for scheduling launches and landings. During storms, operations will rely on redundant power sources and protected, enclosed facilities to safeguard refueling and processing activities.[77]

Finally, the psychological and social well-being of crews must not be overlooked. Extended isolation and confinement in hostile environments can lead to stress, depression, and interpersonal conflict. Crew selection and training will prioritize psychological resilience and teamwork. Habitats will be designed with light therapy, green spaces, and recreational areas to improve morale. Regular Earth connectivity through video calls and digital messaging will help maintain social ties and reduce feelings of isolation.[78] These challenges are substantial but addressable through integrated engineering, operational planning, and human-centered design. By embedding resilience into infrastructure and protocols, spaceports can function as safe, sustainable nodes within a broader interplanetary logistics network. *Table 5* consolidates the 10 highest-priority risks into a structured register, categorized by type and assigned likelihood, impact, and residual risk ratings. Two risks, regolith plume damage and Martian dust storm power interruption, are classified as Critical (High × High) and require dedicated engineering solutions at the earliest design stage.

PHASED DEVELOPMENT APPROACH

Deployment Strategy Overview

The deployment of spaceports on the Moon and Mars requires a structured, phased strategy to ensure operational safety, technical validation, and long-term sustainability. Given the high cost of failure, environmental hazards, and logistical complexity of extraterrestrial operations, an incremental approach, modeled on terrestrial coordination frameworks such as the INCS,[6] is vital.[4] This strategy enables progressive technology maturation, risk mitigation, and adaptive learning, ensuring that infrastructure evolves in alignment with mission requirements and stakeholder capabilities. *Table 6* summarizes the principal milestones, infrastructure targets, and binary go/no-go decision gates for each of the three development phases. The table is intended to serve as a living planning reference for the ISDA as it coordinates contributions from national agencies and commercial partners.

Phase 1: Lunar Prototyping (2030–2040)

This initial phase establishes a robotic prototype spaceport in the lunar south polar region, with a focus on site validation, ISRU demonstration, and autonomous construction. This decade-long effort will validate site selection, test important technologies, and lay the groundwork for future human presence. The primary objectives include confirming the suitability of polar regions, particularly the Shackleton Crater area, for

Table 5. Comparative Risk Register[a]						
Risk	**Category**	**Likelihood**	**Impact**	**Risk Level**	**Mitigation Strategy**	**Residual Risk**
Regolith Plume Damage to Infrastructure	Technical	High	High	Critical	Blast shields, sintered berms, standoff distances	Medium
Dust Storm Power Interruption (Mars)	Environmental	High	High	Critical	Redundant fission + battery storage; hibernation protocol	Low-Medium
Radiation Degradation of Electronics	Technical	High	Medium	High	Rad-hardened components; regolith shielding	Medium
Micrometeoroid Habitat Puncture	Technical	Medium	High	High	Multilayer MMOD shielding; self-sealing membranes	Low
ISRU System Underperformance	Technical	Medium	High	High	Redundant units; Earth-supply contingency	Medium
Communication Blackout (Solar Conjunction)	Operational	Medium	Medium	Medium	Pre-positioning of consumables; autonomous operations protocol	Low
International Governance Failure	Institutional	Low	High	Medium	Phased bilateral agreements; UNOOSA mediation	Medium
Crew Psychological Stress (Isolation)	Human factors	Medium	Medium	Medium	Rotation schedules; real-time psychological support	Low
Launch Vehicle Unavailability	Logistical	Low	High	Medium	Multiprovider contracts; modular manifest planning	Low
Contamination of Scientific Sites	Environmental	Low	Medium	Low	COSPAR planetary protection protocols; exclusion zones	Low

[a]Risk Levels: Critical (High × High), High (High × Medium or Medium × High), Medium, Low.
COSPAR, Committee on Space Research; ISRU, in-situ resource utilization; MMOD, micrometeoroid and orbital debris; UNOOSA, United Nations Office for Outer Space Affairs.

sustained operations, demonstrating the viability of ISRU for producing oxygen, hydrogen, and methane, and proving autonomous construction techniques under real lunar conditions.

Key milestones will begin with robotic scout missions deploying autonomous rovers and drones to conduct high-resolution surveys of candidate sites. These systems will

Table 6. Phased Development Milestones and Key Deliverables			
Milestone	**Phase 1 (2030–2040)**	**Phase 2 (2040–2050)**	**Phase 3 (2050+)**
Primary Objective	Lunar robotic prototype	Martian precursor deployment	Sustained crewed operations
Landing Infrastructure	2 sintered regolith pads (50 m diameter)	3 pads + blast shield berms	Full terminal with 6+ pads
ISRU Output	10 kg/day O_2 (demo scale)	500 kg/day propellant (Mars)	Full closed-loop propellant supply
Power System	10 kWe Kilopower unit	40 kWe reactor array	>100 kWe grid with redundancy
Habitat Capacity	Uninhabited (robotic only)	4-person temporary outpost	12-person permanent crew
Communications Infrastructure	2 lunar relay satellites	Earth-Moon-Mars relay constellation	Full DTN backbone, <1 h latency
Autonomy Level	Supervised autonomous (SAE L3)[79]	Supervised autonomous (SAE L4)[79]	Fully autonomous with human override
Governance Milestone	ISDA charter ratified	Bilateral operational agreements	Multilateral binding framework
Key Go/No-Go Gate	ISRU demo + site stability confirmed	Propellant closure ratio > 0.8	Crew safety certification

DTN, delay-tolerant networking; ISRU, in-situ resource utilization; SAE, Society of Automotive Engineer.

assess regolith composition, radiation levels, thermal cycles, and subsurface ice distribution using LiDAR, multispectral imaging, and ground-penetrating radar. Terrain stability and landing suitability will be evaluated to inform the design of reinforced landing pads.[80] Concurrently, prototype fuel production units will be established to extract water ice from permanently shadowed craters and process it via electrolysis into breathable oxygen and storable hydrogen. Methane synthesis will be tested using imported hydrogen or Earth-based carbon sources, validating closed-loop propellant cycles. Small-scale landing zones will be constructed using sintered regolith or prefabricated materials, equipped with hazard detection systems and precision landing beacons to refine EDL procedures through robotic test landings.[81]

To support future crews, modular habitat prototypes (inflatable or foldable structures) will be deployed and integrated with life-support systems, including air recycling, water purification, and radiation shielding. Their durability will be tested under prolonged exposure to vacuum, thermal extremes, and micrometeorite impacts. A preliminary communication infrastructure will be established through a constellation of relay satellites in lunar orbit, ensuring continuous connectivity with Earth's DSN. Ground-based stations will be set up to manage telemetry, command, and scientific data flow. Comprehensive resource mapping and environmental monitoring will be conducted using distributed sensor arrays to track radiation, thermal shifts, and micrometeorite activity, providing key data for scaling and crewed mission planning.

This phase will involve collaboration among major space agencies (NASA, ESA, and JAXA) and private entities such as SpaceX and Blue Origin, who will contribute transport, robotics, and ISRU technologies. The ISDA will begin forming its core leadership and coordinating international contributions. The expected outcomes include a functional prototype lunar spaceport capable of supporting refueling, cargo handling, and habitat testing; validated ISRU and construction methods; and the establishment of baseline design and operational standards for future expansion.

Phase 2: Martian Precursor Missions (2040–2050)

With the lunar prototype successfully demonstrated, the focus will shift to Mars, preparing for sustainable base establishment and eventual crewed missions. This phase emphasizes autonomous construction, initial infrastructure deployment, and the expansion of logistical and communication networks to support interplanetary coordination.

Robotic systems will be dispatched to construct foundational infrastructure using local materials. Autonomous excavators and 3D printers will fabricate landing pads, hangars, and storage units from processed regolith and basalt composites, minimizing reliance on Earth-launched components.[82] Modular power systems, such as solar farms or compact nuclear reactors like NASA's Kilopower,[72] will be assembled to provide reliable energy. Primary landing infrastructure will be established at equatorial sites such as Gale Crater and Elysium Planitia, chosen for their favorable solar exposure and terrain stability. Dust-mitigation systems will be integrated to protect sensitive equipment during landing operations, and EDL procedures will be tested with large payloads, including simulated crew modules.

Fuel production and storage facilities will expand ISRU capabilities to include atmospheric CO_2 extraction using MOXIE-based approaches.[30] Pilot-scale Sabatier reactors will synthesize methane and water, with cryogenic storage tanks and automated refueling systems enabling spacecraft servicing. A robust communication network will be established through dedicated Martian relay satellites, implementing DTN protocols to manage data transmission across the 4- to 24-min Earth-Mars delay. Local command centers will support autonomous decision-making during communication blackouts. Environmental and health studies will deploy biosensors and radiation monitors to assess surface conditions, while soil and atmospheric analyses will support future agricultural and life-support experiments.

Human habitat prototypes will be delivered to test pressurized shells, internal environmental controls, and closed-loop life-support systems. Underground shelter options, such as lava tubes or excavated tunnels, will be explored for long-term radiation protection.[83] The ISDA will formalize its governance structure and coordinate with UNOOSA, COPUOS, and national agencies, while private firms contribute transport and robotics solutions. Expected outcomes include a functional Martian outpost with basic landing, power, and fuel infrastructure; proven autonomous construction and ISRU-based propellant generation; and a robust communication backbone linking Mars to Earth and orbital assets.

Phase 3: Full-Scale Operations (2050+)

This phase marks the transition from experimental outposts to fully operational spaceports, serving as hubs for scientific research, commercial activity, and interplanetary transit. Lunar and Martian spaceports will support regular launches and landings, with autonomous refueling stations servicing spacecraft such as Starship,[84] cargo haulers, and orbital transfer vehicles. Integrated cargo handling systems will streamline the loading and unloading of supplies, instruments, and mining equipment.

Crewed outposts will house rotating teams of scientists, engineers, and maintenance personnel, supported by permanent habitats and advanced life-support systems. Research laboratories will focus on planetary science, astrobiology, and space medicine, while rovers and drones conduct geological surveys and environmental monitoring. A burgeoning commercial sector will emerge, with mining operations extracting iron, titanium, and platinum-group metals for in-situ construction and export. Tourism ventures will offer short-duration stays for private astronauts, and manufacturing facilities will produce tools, spare parts, and deep-space mission components using local materials.

An interplanetary transport network will be established, connecting Earth, the Moon, and Mars through multinode logistics. Space tugs and orbital transfer platforms will reduce fuel requirements, while AI-optimized scheduling algorithms will coordinate launch windows and mission trajectories. Expansion will extend to the asteroid belt, with scouting missions identifying future spaceport locations and initiating planning for the ABN, which will host fuel depots and manufacturing hubs. The ISDA will coordinate the integration of asteroid-derived resources into the interplanetary supply chain.

Sustainability will be prioritized through carbon neutrality targets, waste recycling, closed-loop life support, and green propulsion. The ISDA will assume full governance, enforcing international treaties and ethical guidelines. Multilateral agreements will ensure equitable access, shared responsibilities, and benefit-sharing among nations and private entities. Educational institutions will expand training programs for off-world engineers, pilots, and technicians. The expected outcomes include a fully integrated interplanetary transportation network, sustainable human settlements on the Moon and Mars, and the emergence of a multiplanetary economy trading in fuel, minerals, data, and services. *Figure 3* presents the integrated development roadmap across all three phases, from initial robotic lunar prototyping through full-scale interplanetary operations, with the ISDA governance and coordination framework spanning the entire timeline. The roadmap should be read alongside *Table 6*, which provides the detailed milestone and go/no-go gate information for each phase.

Optional Extension: ABN Integration (2050+)

As a long-term strategic extension, Phase 3 may incorporate initial deployments within the asteroid belt. Refueling stations will use water ice from asteroids to generate hydrogen and oxygen, creating midway depots between Mars and Jupiter. Mining hubs will extract rare metals for use in Earth-orbit manufacturing or deep-space construction. Scientific platforms on large bodies like Ceres and Vesta will support astrophysics, solar observation, and planetary defense research.[85] These nodes will serve as transit waypoints for missions to Europa, Titan, and beyond, leveraging gravity assists to reduce energy requirements.[86]

The ISDA and ABN will collaborate on joint regulatory frameworks, transportation routes, and refueling protocols. Modular habitats and mining stations will be deployed via reusable cargo vessels and space tugs, with AI-driven logistics systems managing resource flow, mission scheduling, and emergency response across the solar system.

Supporting Frameworks and Digital Infrastructure

Concurrent with physical development, digital and institutional infrastructure will be established. The *SpaceSync* platform, adapted from the proposed INCS model,[6] will use AI to manage launch schedules, cargo tracking, and mission

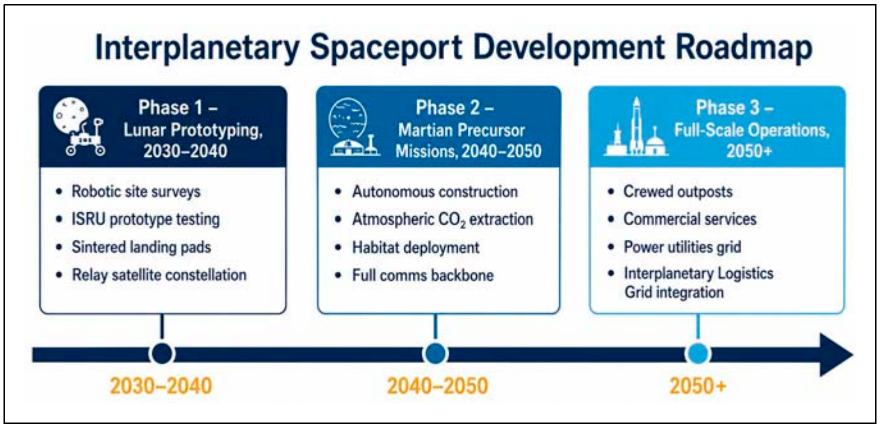


Fig. 3. Interplanetary Spaceport Phased Development Roadmap: Phase 1 Lunar Prototyping (2030–2040), Phase 2 Martian Precursor Missions (2040–2050), and Phase 3 Full-Scale Operations (2050+).

coordination across interplanetary distances. Blockchain-based systems will ensure transparency in fuel usage, material allocation, and financial transactions. Unified command centers on Earth and at key spaceports will oversee mission control, diagnostics, and emergency response.[6]

The ISDA will be formally established as the governing body for interplanetary infrastructure, with international agreements defining ownership, liability, and environmental protections. Training programs will prepare astronauts and technicians through immersive simulations and real-world analog missions.

Phase 4: Interplanetary Expansion (Post-2050)

Looking beyond the Moon and Mars, this forward-looking phase anticipates the next frontier: establishing access points in the outer solar system. Potential developments include fueling stations near icy moons like Europa and Ganymede, spaceports on Titan for methane-based propulsion, and deep-space trade routes leveraging asteroid resources and gravitational assists. Permanent settlements with self-sustaining ecosystems, agriculture, and cultural integration will mark humanity's evolution into a true multiplanetary species. Future work should assess institutional models for standardization, drawing on ICAO and IMO precedents.[2,46]

ECONOMIC AND GOVERNANCE MODEL

Economic and Institutional Framework

The economic and institutional framework for interplanetary spaceports must ensure financial sustainability, regulatory coherence, and equitable participation. It must align with the Outer Space Treaty (1967),[47] and the United Nations Sustainable Development Goals (SDGs),[87] while enabling a self-sustaining multiplanetary economy. Central to this model are PPPs, multilateral governance, diversified funding, and inclusive capacity-building.

Public–Private Partnerships

PPPs are foundational to reducing the financial burden on national governments while harnessing the agility, innovation, and capital of the private sector. In this model, governments play a vital role in funding early-stage R&D, deploying foundational infrastructure, and providing regulatory oversight. National space agencies (such as NASA, ESA, and Roscosmos) will finance high-risk, high-reward missions that may not be economically viable for private firms alone, including lunar fuel production pilots, Martian precursor landings, and the development of autonomous robotics and modular habitat systems. This public investment de-risks technologies and establishes the baseline capabilities upon which commercial ventures can build.

Private sector engagement will be essential for scaling operations and driving innovation. Companies like SpaceX, Blue Origin, and emerging startups will operate refueling services, cargo logistics, space tourism, and resource extraction under licensing agreements with a central authority. These firms will invest in reusable spacecraft, autonomous docking systems, and on-orbit servicing platforms, enhancing mission efficiency and reducing long-term costs. Shared ownership models (such as joint ventures with profit-sharing agreements, cost-recovery mechanisms, and performance-linked incentives) will allow governments and corporations to co-invest in infrastructure, distributing risk and aligning strategic objectives. These partnerships can be formalized through Memoranda of Understanding or multiparty consortium agreements, ensuring accountability and measurable outcomes that contribute to global sustainability goals.

The strategic benefits of this approach are significant. Competition among private firms incentivizes innovation and drives down costs, while modular infrastructure allows for gradual, demand-driven expansion. Furthermore, government-funded R&D can be licensed to private entities, enabling commercialization and broader access to cutting-edge technologies, a process that accelerates the maturation of the space economy.

International Collaboration

Given the inherently global nature of interplanetary exploration, international collaboration is essential for ensuring equitable access, shared responsibility, and efficient resource allocation. Collaborative efforts between established space agencies (NASA, ESA, JAXA, and Roscosmos) and emerging players such as the UAE, India, South Africa, Brazil, and Kenya will promote broad participation and knowledge sharing. These joint ventures can focus on shared technology development, mission planning, and the phased build-out of spaceport infrastructure, with emerging nations benefiting from capacity-building programs and access to advanced operational best practices.

To provide neutral oversight and prevent monopolization by dominant spacefaring nations, the ISDA will serve as an independent governing body. The ISDA will oversee infrastructure siting and design standards, resource allocation and usage rights, safety protocols, environmental protections, and interplanetary traffic coordination. By ensuring equitable access to extraterrestrial resources, the ISDA will uphold the principle that space is the province of all humankind. The authority will align its activities with international law, including the Outer Space Treaty[47] and coordinate closely with the UNOOSA[48] to promote the peaceful and sustainable use of celestial bodies.

A strategic framework will be established to institutionalize cooperation. Regional hubs on Earth (modeled after the proposed INCS[6]) will support lunar and Martian missions, each specializing in different aspects of spaceport development, such as propulsion, habitat design, or robotics. The ISDA will facilitate multilateral agreements among nations to define roles, responsibilities, and benefit-sharing mechanisms. To prevent geopolitical tensions, transparent governance structures and formal dispute resolution mechanisms will be implemented, ensuring that conflicts over resource access or infrastructure control are resolved through dialogue and legal frameworks instead of competition or confrontation.

Real-world precedents offer valuable lessons. The European Space Agency (ESA) demonstrates how multiple nations can collaborate on shared space infrastructure, such as the Guiana Space Center,[88] by pooling expertise and capital. Similarly, the International Civil Aviation Organization (ICAO)[46] provides a proven model for standardizing procedures across borders, a framework the ISDA aims to replicate for interplanetary operations.

Funding Mechanisms

Sustaining interplanetary infrastructure requires a diversified portfolio of funding sources, combining public investment, private capital, and revenue-generating services. Initial funding will come from national space budgets, justified under mandates for scientific exploration, national security, or economic development. International treaties may also establish dedicated funds for interplanetary development, modeled after global initiatives such as the World Bank's Climate Investment Funds[89] or the Global Environment Facility.[90] In addition, advanced concept programs (*e.g.,* NASA's NIAC[91] or ESA's Advanced Concepts Team[92]) will provide seed funding for breakthrough technologies essential to lunar and Martian operations.

As the infrastructure matures, multiple revenue streams will ensure long-term financial sustainability. Satellite refueling services, enabled by ISRU, are expected to become a major source of income, particularly for deep-space missions. Payload transport and logistics (charging fees for cargo delivery, passenger transport, and equipment handling) will generate recurring revenue. Data services, including bandwidth, navigation signals, and telemetry between Earth, the Moon, and Mars, will support a growing digital economy. Power utilities, supplying electricity generated by nuclear reactors or solar arrays to visiting spacecraft, surface vehicles, and third-party operators, represent an additional primary commercial service, particularly given the scarcity and criticality of power in extraterrestrial environments. Royalties on mined resources (such as water ice, rare metals, or regolith-based construction materials) will further diversify income, whether used off-world or returned to Earth. Tourism and research missions will open additional markets, offering commercial access to lunar and Martian facilities for scientists, journalists, and private astronauts. *Table 7* maps each service to its target customer base, revenue model, and projected market maturity, providing a structured basis for financial modeling and investor communication. Power utilities and data relay services are expected to generate early revenue beginning in Phase 1, while crewed accommodation and tourism remain Phase 3 opportunities.

To support this evolving economy, innovative financing tools will be introduced. Green bonds and impact investing will attract environmentally and socially responsible capital to sustainable space development projects.[93] The tokenization of space assets, using blockchain to represent shares in spaceport infrastructure or mining rights, could democratize investment opportunities globally.[6] Governments may also offer tax incentives and subsidy programs to encourage private investment in spaceport-related technologies and operations.

Regulatory Harmonization

To ensure seamless cross-border and interplanetary operations, regulatory frameworks must evolve in parallel with technological and financial models. The ISDA will lead the development of unified licensing protocols for landing rights, resource extraction, and habitation permits, drawing inspiration from ICAO-style aviation certification.[46] A standardized liability and insurance regime will be established, requiring mandatory coverage for spaceport operators and defining clear procedures for claim settlement in the event of launch failures or accidents.

Environmental protection will be a core principle. Inspired by terrestrial environmental laws, the ISDA will enforce strict guidelines on waste management, contamination prevention, and the preservation of planetary heritage sites, for example, Apollo landing zones or scientifically significant Martian formations. An intellectual property framework will be developed to manage patents, technology transfer, and open-source contributions, fostering innovation while preventing monopolies and ensuring fair competition. *Table 8* catalogs the eight principal legal instruments and governance frameworks directly relevant to interplanetary spaceport operations. Notably, the Artemis Accords (2020)[97] had over 50 signatories by the end of 2025 and represent the most operationally specific

Table 7. Commercial Revenue Streams and Target Markets				
Service	Target Customers	Revenue Model	Market Maturity	Phase Available
Propellant Supply (ISRU-Derived)	Deep-space mission operators, commercial launchers	Per-kg dispensed	Emerging	Phase 2+
Power Utilities (Electricity Supply)	Visiting spacecraft, surface operators, third parties	Per-kWh metered	Nascent	Phase 1+
Cargo Handling and Logistics	NASA, ESA, commercial payload operators	Per-kg manifest fee	Emerging	Phase 2+
Data Relay and Communications	Satellite operators, mission teams	Bandwidth subscription	Near-term	Phase 1+
Scientific Facility Access	Universities, national agencies, private researchers	Time-based access fee	Near-term	Phase 2+
Crew Accommodation and Life Support	Government astronauts, commercial crew	Per-person-day rate	Future	Phase 3
Tourism and Media Access	Private individuals, broadcasters	Premium per-visit fee	Speculative	Phase 3
Mined Resource Royalties	Space mining companies, Earth return operators	Revenue-share/royalty	Speculative	Phase 3
In-Orbit Servicing Staging	Satellite servicing companies	Berthing + service fee	Emerging	Phase 2+

ISRU, in-situ resource utilization.

bilateral framework, explicitly addressing ISRU, safety zones, and interoperability, three priorities central to the ISDA's mandate.

Capacity Building

Ensuring that emerging economies can participate in the interplanetary future is key for long-term stability, inclusivity,

Table 8. International Legal and Governance Framework				
Instrument	Year	Status	Relevance to Interplanetary Spaceports	Key Provision
Outer Space Treaty[47]	1967	114 state parties	Primary governing framework	No national sovereignty; state responsibility for national activities
Rescue Agreement[94]	1968	98 state parties	Crew safety obligations	Duty to rescue and return astronauts
Liability Convention[95]	1972	98 state parties	Infrastructure damage liability	Launching state liable for damage caused by space objects
Registration Convention[96]	1976	72 state parties	Spaceport asset identification	Objects must be registered with UN
Artemis Accords[97]	2020	50+ signatories (2025)	Bilateral operational framework	ISRU permissibility; safety zones; interoperability norms
COSPAR Planetary Protection Policy[98]	1967 (Updated)	Voluntary/widely adopted	Environmental protection	Contamination prevention for scientifically sensitive sites
ITU Radio Regulations[99]	Ongoing	Binding for ITU members	Communications spectrum allocation	Frequency coordination for relay satellites
National Space Laws (*e.g.*, US SPACE Act 2015,[100] 2017 Luxembourg Space Resources Law[101])	Various	Binding nationally	Resource extraction rights	Allow nationals to own extracted resources

COSPAR, Committee on Space Research; ISRU, in-situ resource utilization; ITU, International Telecommunication Union; UN, United Nations.

and innovation. To this end, the ISDA will establish an Emerging Nations Fund (ENF) to support spaceport-related education, technical training, and small-scale infrastructure development in low-income countries. STEM and technical training programs will be launched in partnership with universities and technical institutions, equipping engineers, technicians, and mission planners with the skills needed for extraterrestrial logistics and operations. Affordable access programs will offer reduced rates or subsidized entry to spaceport services for nations and organizations from developing regions, lowering the barrier to participation in the space economy. These initiatives are expected to yield significant outcomes: increased diversity among space actors, stronger global partnerships, and greater innovation through localized problem-solving and cross-cultural collaboration.

Resilience and Redundancy

As Earth-based spaceports must prepare for natural disasters and geopolitical instability, interplanetary facilities must plan for systemic risks unique to extraterrestrial environments. Mission continuity protocols will ensure that if one spaceport becomes temporarily inoperable (due to a Martian dust storm or a lunar radiation event), alternate routes and backup nodes can maintain operations. Redundant communication and navigation systems, including multiple relay satellites and ground stations, will preserve connectivity during blackouts or equipment failures. Regular emergency response drills, involving the ISDA, UNOOSA, and private operators, will test contingency plans and improve readiness for crises ranging from habitat breaches to supply chain disruptions.

Future Economic Integration

As spaceports mature, they will evolve from scientific outposts into commercial hubs supporting a growing interplanetary economy. Key economic opportunities will include the production and trade of fuel, with lunar and Martian refueling stations serving as vital waypoints for deep-space missions. Raw materials (*e.g.,* rare metals from asteroids or regolith-based construction supplies) will be exported to Earth or used in orbit for in-space manufacturing. Microgravity environments will enable the production of high-value goods, including ultra-pure pharmaceuticals and advanced optical fibers, which cannot be efficiently manufactured on Earth. Scientific and cultural tourism will also expand, supporting short-term stays for researchers, artists, and private citizens, thereby generating additional revenue.

Interplanetary trade routes will form the backbone of this economy. The Earth-Moon-Mars Corridor will serve as the primary transit lane, with asteroid belt nodes acting as refueling and maintenance stops. The ABN will leverage mining operations to supply raw materials and fuel, integrating seamlessly with the broader logistics network.[54]

Strategic Implementation Roadmap

To ensure long-term success, the economic and governance model will follow a phased rollout strategy, mirroring the approach of the proposed INCS framework.[6] Phase 1 will focus on consortium formation and feasibility analysis, identifying key stakeholders and securing initial funding through PPPs and multilateral agreements. Phase 2 will establish pilot spaceport nodes on the Moon, linking them with terrestrial proposed INCS-participating spaceports to test governance models, regulatory harmonization, and disaster resilience. Phase 3 will scale the network to include additional lunar and Martian locations, implementing AI-driven scheduling platforms, blockchain-based transaction systems, and predictive analytics to optimize operations. Finally, Phase 4 will formalize the ISDA's governance structure and integrate it with COPUOS, embedding long-term sustainability goals such as carbon neutrality and waste reduction into its mandate. This comprehensive model provides a pragmatic, evidence-based pathway toward a resilient, interoperable, and inclusive interplanetary infrastructure system.

CONCLUSION

The development of interplanetary spaceports on the Moon and Mars constitutes a key step in enabling sustained human presence beyond Earth. These facilities will serve as multifunctional nodes supporting scientific research, crewed missions, cargo logistics, ISRU, and interplanetary transportation. Instead of functioning as isolated launch and landing zones, they are intended to operate as integrated components of a broader interplanetary logistics network, facilitating both governmental and commercial activities across the inner solar system.

This paper outlines a technically grounded architecture for lunar and Martian spaceports, integrating ISRU, autonomy, and modular design to address environmental and logistical constraints. Important gaps remain in materials validation, power resilience, and interoperability standards. Future work should prioritize Earth-analog demonstrations and open-architecture interface protocols to enable multistakeholder participation. By anchoring development in empirical data and COSPAR-aligned sustainability principles, this framework supports the transition from exploration to sustained operations beyond Earth orbit.

The deployment of interplanetary spaceports thus represents an engineering and logistical undertaking but also a governance and policy challenge. Its success will depend on the ability of the international community to align technical development with principles of sustainability, inclusivity, and shared stewardship. By grounding spaceport development in empirical validation, multilateral coordination, and phased implementation, humanity can establish a resilient and interoperable infrastructure capable of supporting sustained operations across multiple celestial bodies. This effort marks a necessary evolution in space activity, from episodic missions to enduring, system-level infrastructure. The decisions made in the coming decade will shape the operational, economic, and ethical foundations of humanity's multiplanetary future—one spaceport at a time.

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AUTHOR'S CONTRIBUTIONS

The author declares that this work represents original research. The author conceived the research idea, developed the conceptual framework, performed all literature reviews, carried out technical and policy analyses, designed the proposed architecture, and wrote the entire article. All data, arguments, and conclusions presented are the author's own, unless explicitly cited. This article has not been published previously and is not under consideration for publication elsewhere. The author takes full responsibility for the integrity, accuracy, and scholarly rigor of this work.

DISCLOSURE

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REFERENCES

1. Mazur SK, Minich D, Szocik K. Legal, political, and ethical challenges of Mars settlement. Astropolitics 2020;18(3):223–237; doi: 10.1080/14777622.2021.1907994
2. National Aeronautics and Space Administration. Moon to Mars architecture. NASA; 2025. Available from: https://www.nasa.gov/moontomarsarchitecture/ [Last accessed: May 13, 2026].
3. Sanders GB, Larson WE. Integration of in-situ resource utilization into Lunar/Mars exploration through field analogs. Adv Space Res 2011;47(1):20–29; doi: 10.1016/j.asr.2010.08.020
4. Drake BG, et al. Human exploration of Mars, design reference architecture 5.0. In: IEEE Xplore. 2010; pp. 1–24, doi: 10.1109/AERO.2010.5446736
5. Zhang R, Wang T, Wang G. Review of Lunar regolith forming technologies for in-situ manufacturing/construction on the Lunar surface. Addit Manuf Front 2025;4(3):200238; doi: 10.1016/j.amf.2025.200238
6. Kanjumba WC. An integrated network for commercial spaceports: Enhancing global space access and collaboration—paper ID: 95293. International Astronautical Federation; 2025. Available from: https://dl.iafastro.directory/event/IAC-2025/paper/95293/ [Last accessed: May 13, 2026].
7. Crawford IA. Lunar resources: A review. Prog Phys Geogr 2015;39(2):137–167.
8. Zuber MT, Head JW, Smith DE, et al. Constraints on the volatile distribution within Shackleton crater at the Lunar South pole. Nature 2012;486(7403):378–381; doi: 10.1038/nature11216
9. Speyerer EJ, Robinson MS. Persistently illuminated regions at the Lunar poles: Ideal sites for future exploration. Icarus 2013;222(1):122–136; doi: 10.1016/j.icarus.2012.10.010
10. Schrunk D, Sharpe BL, Cooper BL, Thangavelu M. The Moon: Resources, Future Development, and Settlement. 2nd ed. *Springer Praxis Books*: Chichester, UK; 2007.
11. Sharpe BL, Schrunk DG. Malapert mountain: Gateway to the Moon. Adv Space Res 2003;31(11):2467–2472; doi: 10.1016/s0273-1177(03)00535-0
12. Luchsinger KM, Chanover NJ, Strycker PD, et al. Water within a permanently shadowed lunar crater: Further LCROSS modeling and analysis. Icarus 2021;354:114089; doi: 10.1016/j.icarus.2020.114089
13. Washington University. Lunar orbital data explorer - product search. Washington University in St. Louis; 2026. Available from: https://ode.rsl.wustl.edu/moon/productsearch [Last accessed: May 13, 2026].
14. Coyan J, Siegler M, Martinez-Comacho J, et al. Prospectivity modeling of the NASA VIPER landing site at Mons Mouton near the Lunar South pole. Planet Sci J 2025;6(5):105; doi: 10.3847/psj/adbc6c
15. Culbert CJ, Ehlmann BL, Fraeman AA. Revolutionizing access to the mars surface a strategy: Frequent. Affordable. Bold. W. M. Keck Institute for Space Studies (KISS); 2024. Available from: https://www.kiss.caltech.edu/ [Last accessed: May 13, 2026].
16. Hintze P, Meier A, Shah M, Devor R. Sabatier system design study for a Mars ISRU propellant production plant. NASA Technical Reports Server; 2018. Available from: https://ntrs.nasa.gov/citations/20180004697 [Last accessed: May 13, 2026].
17. Mouginis-Mark PJ. Olympus mons volcano, Mars: A photogeologic view and new insights. Geochemistry 2018;78(4):397–431; doi: 10.1016/j.chemer.2017.11.006
18. Peulvast J-P, Mège D, Chiciak J, et al. Morphology, evolution and tectonics of Valles Marineris Wallslopes (Mars). Geomorphology 2001;37(3–4):329–352; doi: 10.1016/s0169-555x(00)00085-4

19. Buckner DK, Mojarro A. An organics-forward approach to searching for life on Mars. Nat Commun 2026;17(1); doi: 10.1038/s41467-026-72266-2
20. Grotzinger JP, Crisp JA, Vasavada AR. Curiosity's mission of exploration at Gale crater, Mars. Elements 2015;11(1):19–26; doi: 10.2113/gselements.11.1.19
21. Banerdt WB, Smrekar SE, Lognonné P, et al. Initial results from the InSight mission on Mars. Nat Geosci 2020;13(3):183–189; doi: 10.1038/s41561-020-0544-y
22. Bramson AM, Byrne S, Bapst J, et al. Preservation of Midlatitude ice sheets on Mars. JGR Planets 2017;122(11):2250–2266; doi: 10.1002/2017je005357
23. National Aeronautics and Space Administration. Mars 2020: Perseverance rover. NASA Science; 2025. Available from: science.nasa.gov/mission/mars-2020-perseverance/ [Last accessed: May 13, 2026].
24. Zurek RW, Smrekar SE. An overview of the Mars Reconnaissance Orbiter (MRO) science mission. J Geophys Res 2007;112(E5); doi: 10.1029/2006je002701
25. Planetary Atmospheres Node of the Planetary Data System. Mars reconnaissance orbiter archive—Mars Climate Sounder (MCS): Temperature profiles. New Mexico State University; 2025. Available from: pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/MARS/atmosphere_temp_prof.html [Last accessed: May 13, 2026].
26. National Aeronautics and Space Administration. Mars exploration program landing sites. NASA; 2011. Available from: marsoweb.nas.nasa.gov/landingsites/landingsites.html [Last accessed: May 13, 2026].
27. National Aeronautics and Space Administration. 2020 NASA technology taxonomy. NASA; 2023. Available from: www.nasa.gov/otps/2020-nasa-technology-taxonomy/ [Last accessed: May 13, 2026].
28. European Cooperation for Space Standardization. ECSS-E-HB-11A – technology readiness level (TRL) guidelines (1 March 2017). ECSS; 2017. Available from: https://ecss.nl/home/ecss-e-hb-11a-technology-readiness-level-trl-guidelines-1-march-2017/ [Last accessed: May 13, 2026].
29. Sowers GF, Dreyer CB. Ice mining in Lunar permanently shadowed regions. New Space 2019;7(4):235–244; doi: 10.1089/space.2019.0002
30. Hoffman JA, Hecht MH, Rapp D, et al. Mars oxygen ISRU experiment (MOXIE)—preparing for human Mars exploration. Sci Adv 2022;8(35):eabp8636; doi: 10.1126/sciadv.abp8636
31. Yang Z, Liu Y, Wang H, et al. Making fuels on Mars by Sabatier reaction. Innov Energy 2025;2(2):100086; doi: 10.59717/j.xinn-energy.2025.100086
32. Notardonato WU, Swanger AM, Fesmire JE, et al. Zero boil-off methods for large-scale liquid hydrogen tanks using integrated refrigeration and storage. IOP Conf Ser: Mater Sci Eng 2017;278:e012012; doi: 10.1088/1757-899x/278/1/012012
33. Johansen MR. An update on NASA's Lunar dust mitigation strategy. NASA Technical Reports Server; 2020. Available from: https://ntrs.nasa.gov/ [Last accessed: May 13, 2026].
34. Slaba TC, Bahadori AA, Reddell BD, et al. Optimal shielding thickness for Galactic Cosmic ray environments. Life Sci Space Res (Amst) 2017;12:1–15; doi: 10.1016/j.lssr.2016.12.003
35. Bao C, Zhang D, Wang Q, et al. Lunar in situ large-scale construction: Quantitative evaluation of regolith solidification techniques. Engineering 2024;39:204–221; doi: 10.1016/j.eng.2024.03.004
36. Beaty D, Albee A, Allen CC, et al. Analysis of the Mars feedforward value of possible Lunar objectives. Mars Forward Lunar Objectives Science Analysis Group (MFLO-SAG); 2007. Available from: https://www.lpi.usra.edu/ [Last accessed: May 13, 2026].
37. Haberle RM, et al. The Atmosphere and Climate of Mars. Cambridge University Press; 2017.
38. Cesaretti G, Dini E, De Kestelier X, et al. Building components for an outpost on the Lunar soil by means of a novel 3D printing technology. Acta Astronaut 2014;93:430–450; doi: 10.1016/j.actaastro.2013.07.034
39. National Aeronautics and Space Administration. Lunar Communications Relay Demonstration (LCRD). NASA; 2025. Available from: https://www.nasa.gov/mission/laser-communications-relay/ [Last accessed: May 13, 2026].
40. National Aeronautics and Space Administration. Mars reconnaissance orbiter. NASA Science; 2025. Available from: https://science.nasa.gov/mission/mars-reconnaissance-orbiter/ [Last accessed: May 13, 2026].
41. National Aeronautics and Space Administration. 'Seeing' the Earth, Moon, and Sun to scale. NASA; 2000. Available from: https://www.grc.nasa.gov/www/k-12/Numbers/Math/Mathematical_Thinking/seeing_the_earth_moon.htm [Last accessed: May 13, 2026].
42. Bhasin KB, Hayden JL. Space internet architectures and technologies for NASA enterprises. IEEE Aerosp Conf 2001; doi: 10.1109/aero.2001.931275
43. National Aeronautics and Space Administration. Delay/disruption tolerant networking. NASA; 2025. Available from: https://www.nasa.gov/communicating-with-missions/delay-disruption-tolerant-networking/ [Last accessed: May 13, 2026].
44. National Aeronautics and Space Administration. Deep space network. NASA; 2025. Available from: https://www.nasa.gov/communicating-with-missions/dsn/ [Last accessed: May 13, 2026].
45. Bogue R. Robots for space exploration. Ind Rob: Int J 2012;39(4):323–328; doi: 10.1108/01439911211227872
46. International Civil Aviation Organization. Convention on international civil aviation—doc 7300. ICAO; 2025. Available from: www.icao.int/publications/doc-series/convention-international-civil-aviation-doc-7300 [Last accessed: May 13, 2026].
47. United Nations Office for Outer Space Affairs. The outer space treaty. UNOOSA; 1967. Available from: https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introouterspacetreaty.html [Last accessed: May 13, 2026].
48. United Nations Office for Outer Space Affairs. Space law treaties and principles. UNOOSA; 2022. Available from: https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties.html [Last accessed: May 13, 2026].
49. United Nations Committee on the Peaceful Uses of Outer Space. Committee on the peaceful uses of outer space: Membership evolution. UNOOSA; 2024. Available from: https://www.unoosa.org/oosa/de/ourwork/copuos/members/evolution.html [Last accessed: May 13, 2026].
50. Kanjumba WC. Omega spaceport: Establishing Kenya as a global hub for equatorial space access and commercial space sector growth—paper ID: 95292. International Astronautical Federation; 2025. Available from: https://dl.iafastro.directory/event/IAC-2025/paper/95292/ [Last accessed: May 13, 2026].
51. National Aeronautics and Space Administration. Surface systems. NASA; 2025. Available from: www.nasa.gov/reference/marshall-moon-mars-surface-technologies-and-systems/ [Last accessed: May 13, 2026].
52. European Space Agency. Living in space. ESA; 2024. Available from: www.esa.int/Enabling_Support/Preparing_for_the_Future/Discovery_and_Preparation/Living_in_space [Last accessed: May 13, 2026].
53. Heinicke C, Arnhof M. A review of existing analog habitats and lessons for future Lunar and Martian habitats. REACH 2021;21–22:100038; doi: 10.1016/j.reach.2021.100038
54. Kanjumba WC. Integrating technology, sustainability, and human resilience under a fully functional space ecosystem. In: COSPAR Symposium; 2025.
55. National Aeronautics and Space Administration. Why space radiation matters. NASA; 2017. Available from: https://www.nasa.gov/missions/analog-field-testing/why-space-radiation-matters/ [Last accessed: May 13, 2026].
56. National Aeronautics and Space Administration. Solar wind on the Moon. NASA Science; 2025. Available from: https://science.nasa.gov/moon/solar-wind/ [Last accessed: May 13, 2026].
57. Dobney W, Mols L, Mistry D, et al. Evaluation of deep space exploration risks and mitigations against radiation and microgravity. Front Nucl Med 2023;3:1225034; doi: 10.3389/fnume.2023.1225034
58. Blunier S, Athanasopoulos D, Dittmar C, et al. Regolith-based Lunar habitat for astronaut radiation protection and organ dose assessment. Planet Space Sci 2025;261:106116; doi: 10.1016/j.pss.2025.106116
59. Feng Y, Pan PZ, Tang X, et al. A comprehensive review of Lunar lava tube base construction and field research on a potential Earth test site. Int J Min Sci Technol 2024;34(9):1201–1216; doi: 10.1016/j.ijmst.2024.06.003
60. Ferrone K, Willis C, Guan F, et al. A review of magnetic shielding technology for space radiation. Radiation 2023;3(1):46–57; doi: 10.3390/radiation3010005
61. Catling DC. Mars atmosphere: history and surface interactions. In: Encyclopedia of the Solar System, Third Edition. Elsevier; 2014; pp. 343–357, doi: 10.1016/b978-0-12-415845-0.00016-5
62. Ehresmann B, Zeitlin C, Hassler DM, et al. The Martian surface radiation environment at solar minimum measured with MSL/RAD. Icarus 2023;393:115035; doi: 10.1016/j.icarus.2022.115035

63. Townsend LW. Overview of active methods for shielding spacecraft from energetic space radiation. Physica Medica 2001;17(Suppl 1):1–5. Available from: https://pubmed.ncbi.nlm.nih.gov/
64. Hassler DM, Zeitlin C, Wimmer-Schweingruber RF, et al.; MSL Science Team. Mars' surface radiation environment measured with the Mars science laboratory's curiosity rover. Science 2014;343(6169):1244797; doi: 10.1126/science.1244797
65. Iwase S, Nishimura N, Tanaka K, Mano T. Effects of microgravity on human physiology. In: Beyond LEO - Human Health Issues for Deep Space Exploration. IntechOpen: London, UK; 2020; doi: 10.5772/intechopen.90700
66. National Aeronautics and Space Administration. Exercise countermeasures. NASA; 2024. Available from: https://www.nasa.gov/reference/jsc-exercise-countermeasures/ [Last accessed: May 13, 2026].
67. NASA Technology Transfer Program. Spacecraft with artificial gravity modules. NASA Technology Transfer Program; 2025. Available from: https://technology.nasa.gov/ [Last accessed: May 13, 2026].
68. Marquez J, Kaiser MK, Feinberg AN, et al. Evaluation of human and automation/robotics integration needs for future human exploration missions. NASA Ames Human Systems Integration Division. Available from: https://human-factors.arc.nasa.gov/ [Last accessed: May 13, 2026].
69. Lei S, Guoqing Z, Yaohui W, et al. Strategies and prospects for energy storage in future Lunar base. Adv Mech Eng 2025;17(8); doi: 10.1177/16878132251362316
70. Kerslake TW, Gefert LP. Solar power system analyses for electric propulsion missions. In: Intersociety Energy Conversion Engineering Conference. NASA/TM-1999-209289. 1999; doi: 10.1109/62.894171
71. Kumba K, Upender P, Buduma P, et al. Solar tracking systems: Advancements, challenges, and future directions: A review. Energy Reports 2024;12:3566–3583; doi: 10.1016/j.egyr.2024.09.038
72. National Aeronautics and Space Administration. Kilopower. NASA; 2017. Available from: https://www.nasa.gov/directorates/stmd/tech-demo-missions-program/kilopower-hmqzw/ [Last accessed: May 13, 2026].
73. Patel MR, Beik O. Spacecraft Power Systems. CRC Press; 2023, doi: 10.1201/9781003344605
74. Zanon P, Dunn M, Brooks G. Current Lunar dust mitigation techniques and future directions. Acta Astronaut 2023;213:627–644; doi: 10.1016/j.actaastro.2023.09.031
75. Forget F, Hourdin F, Fournier R, et al. Improved general circulation models of the martian atmosphere from the surface to above 80 km. J Geophys Res 1999;104(E10):24155–24175; doi: 10.1029/1999JE001025
76. National Aeronautics and Space Administration. NASA passive thermal control engineering guidebook. NASA Technical Reports Server (NTRS); 2023. Available from: https://ntrs.nasa.gov/ [Last accessed: May 13, 2026].
77. Kahre MA, et al. The Mars dust cycle. The Atmosphere and Climate of Mars, Cambridge University Press; 2017; pp. 295–337, doi: 10.1017/9781139060172.010
78. Kanas N, Manzey D. Space Psychology and Psychiatry. Springer: New York, NY; 2008; doi: 10.1007/978-1-4020-6770-9
79. SAE International. J3016_202104 - taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles. SAE; 2021. Available from: https://www.sae.org/standards/j3016_202104-taxonomy-definitions-terms-related-driving-automation-systems-road-motor-vehicles [Last accessed: May 13, 2026].
80. Jia B, Fa W, Xie M, et al. Regolith properties in the Chang'E-5 landing region of the Moon: Results from multi-source remote sensing observations. JGR Planets 2021;126(7); doi: 10.1029/2021JE006934
81. Zhang P, Wu Q, Chen Y, et al. Overview of the lunar in-situ resource utilization techniques for future Lunar missions. Space 2023;3; doi: 10.34133/space.0037
82. Obosu M, Frimpong S. Advances in automation and robotics: The state of the emerging future mining industry. J Saf Sustain 2025;2(3):181–194; doi: 10.1016/j.jsasus.2025.05.003
83. Sauro F, Pozzobon R, Massironi M, et al. Lava tubes on Earth, Moon and Mars: A review on their size and morphology revealed by comparative planetology. Earth Sci Rev 2020;209:103288; doi: 10.1016/j.earscirev.2020.103288
84. SpaceX. Starship. SpaceX; 2025. Available from: https://www.spacex.com/ [Last accessed: May 13, 2026].
85. Rayman MD. Lessons from the Dawn mission to Ceres and Vesta. Acta Astronaut 2020;176:233–237; doi: 10.1016/j.actaastro.2020.06.023
86. National Air and Space Museum. Gravity assist. Smithsonian's National Air and Space Museum. Available from: https://timeandnavigation.si.edu/ [Last accessed: May 13, 2026].
87. United Nations. Transforming our world: The 2030 agenda for sustainable development. United Nations; 2025. Available from: https://sdgs.un.org/ [Last accessed: May 13, 2026].
88. Center for Strategic and International Studies. Spaceports of the world. Aerospace Security; 2023. Available from: https://aerospace.csis.org/data/spaceports-of-the-world/ [Last accessed: May 13, 2026].
89. World Bank. Climate Investment Funds (CIF). CIF; 2025. Available from: https://www.cif.org/ [Last accessed: May 13, 2026].
90. Global Environment Facility. Global environment facility. GEF; 2025. Available from: https://www.thegef.org/ [Last accessed: May 13, 2026].
91. National Aeronautics and Space Administration. NASA innovative advanced concepts. NASA; 2025. Available from: https://www.nasa.gov/ [Last accessed: May 13, 2026].
92. European Space Agency. Advanced concepts team. ESA; 2025. Available from: https://www.esa.int/ [Last accessed: May 13, 2026].
93. International Finance Corporation. Creating green bond markets—insights, innovations, and tools from emerging markets. World Bank Group; 2018. Available from: https://www.climatebonds.net/ [Last accessed: May 13, 2026].
94. United Nations Office for Outer Space Affairs. Rescue agreement. UNOOSA; 1968. Available from: https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/rescueagreement.html [Last accessed: May 13, 2026].
95. United Nations Office for Outer Space Affairs. Liability convention. UNOOSA; 1972. Available from: https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introliability-convention.html [Last accessed: May 13, 2026].
96. United Nations Office for Outer Space Affairs. Registration convention. UNOOSA; 1976. Available from: https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/registration-convention.html [Last accessed: May 13, 2026].
97. National Aeronautics and Space Administration. Artemis accords. NASA; 2026. Available from: https://www.nasa.gov/artemis-accords/ [Last accessed: May 13, 2026].
98. Committee on Space Research. COSPAR policy on planetary protection. COSPAR Business; 2024, doi: 10.1016/j.lssr.2022.12.001
99. International Telecommunication Union. Radio regulations. ITU; 2024. Available from: https://www.itu.int/pub/R-REG-RR [Last accessed: May 13, 2026].
100. U.S. Congress. H.R.2262 - U.S. commercial space launch competitiveness act. Congress; 2015. Available from: www.congress.gov/bill/114th-congress/house-bill/2262 [Last accessed: May 13, 2026].
101. Official Journal of the Grand Duchy of Luxembourg. Law of July 20th 2017 on the exploration and use of space resources. Official Journal of the Grand Duchy of Luxembourg; 2017. Available from: https://legilux.public.lu/eli/etat/leg/loi/2017/07/20/a674/jo/en [Last accessed: May 13, 2026].

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